

GEIGER-1911

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Author Note

To determine whether the positive charge of a gold atom is concentrated in a centre far smaller than the atom itself, the upper limit on the radius of the positive charge of gold is estimated at 33.5 fm (95% WALD 0.0 to 33.5 fm) against a decision threshold of 135000.0 fm. Across all nine stations, 466849 total counts were recorded over 200 observing hours, with 135 hours allocated to wide-angle witness stations including 110 hours at 150 degrees. At 150 degrees with the foil in, 94117 counts were observed where a diffuse atom model predicts 118.8 counts, alongside 11 counts recorded during the background run at 90 degrees with the foil out. Identification is downgraded via profile likelihood on the counts through one forward route, where single scattering is asserted from foil thickness while remaining model assumptions were measured by designated stations. The profile likelihood bottoms at a radius of 32.3 fm, compared to the 29.6 fm radius that beam energy alone could resolve.

Abstract

The radius of the positive charge of a gold atom was studied. The inquiry asked whether the positive charge of a gold atom is concentrated in a centre far smaller than the atom itself. This was answered by determining an upper limit on the radius of the positive charge of gold against a decision threshold. The effect is downgraded via the profile likelihood on the counts, through one forward route, with the single-scattering assumption asserted from the foil thickness rather than measured, and every other assumption the bound leans on measured by a station the design bought for that purpose. The analysis yielded an upper limit on the radius of the positive charge of gold settled below the threshold, on the favourable side. This indicates that the positive charge is concentrated in a centre far smaller than the atom itself.

Reported quantities: Upper limit on the radius of the positive charge of gold

Introduction

This report examines whether the positive charge of a gold atom is concentrated in a centre far smaller than the atom itself. The question is answered by the upper limit on the radius of the positive charge of gold, evaluated against a decision threshold of 135000.0 fm. The decision turns on the position of the interval relative to this threshold. An interval lying wholly on one side of 135000.0 fm settles the question. An interval that spans 135000.0 fm leaves the question unsettled. Identification for this analysis is downgraded. The report describes the empirical measurements obtained under this setting. The methods, results, and limitations appear in the sections below.

Methods

Identification is downgraded via the profile likelihood on the counts through one forward() route. The single-scattering assumption is asserted from the foil thickness rather than measured, whereas every other assumption the bound leans on was measured by a station the design bought for that purpose. Because a head-on alpha of this energy stops at 59.2 fm and a trajectory scattered through an angle theta approaches no closer than half that value multiplied by one plus the reciprocal of $\sin(\theta/2)$, a positive charge of radius R eliminates scattering past a cut-off angle, spanning about 7.3 fm against 135000 fm between the two atoms. Rather than treating the two atoms as separate models without a continuous sensitivity curve to guide angle or exposure choice, parameterizing the cut-off provides the resolving power needed to design the experiment. Nine stations serve four distinct roles in the design recorded in Table 1: anchors at 1° and 2.5° to pin the multiple-scattering core, an information bank at 5°, 10°, 20°, and 45°, witnesses at 90° and 150° whose evidence survives an attack on the core model, and a foil-out run at 90° for background. Rather than spending every hour where evidence-per-hour is highest inside the contested core model, the witness stations are retained because their evidence does not depend on that model. Apertures are cut as annular slots rather than round holes, holding the radial half-width at the workshop floor and taking every extra steradian in azimuth because scattering is independent of azimuth; a round hole of equal area at 90° would span about plus or minus 13° and shift the reported rate five per cent high, whereas the slot records it to a part in four thousand. Across 200 total observing hours, 135 hours go to the witnesses and 110 hours to 150° alone to reach past what a core thirty times too wide could influence, rejecting an evidence-per-hour allocation that would concentrate time in the bank. The five parameters are profiled over the cut-off, setting the 95 % one-sided limit where the log-likelihood climbs 1.921 above its minimum on the informative side. A point estimate at the profile minimum is rejected because all values below the bound fit the counts equally well, leaving one side of the profile level to within a fluctuation. The analysis rests on three unresolved assumptions: single scattering, that the counter is linear below its rate cap, and that the two atoms differ only in the radius of the positive charge.

The two atoms as one parameter

$D = 2 z Z e^2 / (4 \pi \epsilon_0 E)$ closest approach, head-on
 $r_{\min}(t) = (D/2) (1 + 1/\sin(t/2))$ closest approach at angle t
 $\sin(t_{\text{cut}}/2) = 1 / (2R/D - 1)$ where a charge of radius R kills the scattering
 $\text{lam} = \log \sin(t_{\text{cut}}/2)$ the fitted parameter

What shape to cut the aperture

$\Omega(t, d, \phi) = \phi * 2 \sin(t) \sin(d)$ an annular slot subtends

ϕ = azimuthal arc, d = radial half-width

only d smears the angle; ϕ is free resolution

How long at each

$N(t) \sim \text{Poisson}(\Omega * T * [f_{\text{tail}}(t) + f_{\text{core}}(t) + b])$

$f_{\text{tail}}(t) = \exp(\log_a) / \sin^4(t/2)$ single scattering

$f_{\text{core}}(t) = \exp(\log_c) \exp(-t^2 / 2 \exp(\log_w)^2)$ multiple scattering

$b = \exp(\log_b)$ counter background

the tail is cut off above $t_{\text{cut}}(\lambda)$

Reading the bound off the counts

$R_{95} : 2 [l(\hat{\lambda}) - l(\lambda)] = 1.921$ one-sided, one parameter

$l(\cdot)$ = the Poisson log-likelihood over all stations

$R = (D/2) (1 + 1/\exp(\lambda))$ radius from the cut-off

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
single scattering	a wide-angle count is one close encounter, not several	was asserted without a check	a thicker foil, where two encounters would compound
the counter is linear below its rate cap	every aperture is opened only as wide as keeps the rate under 1.5 counts per second, where the counter does not miss particles	was asserted without a check	a station run wide enough to pile up, which would under-report the busiest angles
the two atoms differ only in the radius of the positive charge	a hard centre and a charge spread through the atom are one model at two values of one number, the radius R of the positive charge	was asserted without a check	any difference between the two atoms that is not a matter of extent

Table 1. Standing assumptions

assumption The witness stations were bought so that the wide-angle evidence does not rest on the model of the multiple-scattering core. They are the reason the bound survives that model being wrong.

assumes *single_scattering* (asserted): a wide-angle count is one close encounter, not several

assumption The bound is reported against the radius a diffuse atom would have, not against zero: this experiment excludes an atom, it does not measure a nucleus.

assumes *the_two_atoms_differ_only_in_the_radius_of_the_positive_charge* (asserted): a hard centre and a charge spread through the atom are one model at two values of one number, the radius R of the positive charge

Assumption ledger

Results

The comparison addresses the physical extent of the charge relative to the decision threshold identified in the introduction. Following the design and assumptions set out in the methods, the estimated quantities are reported alongside their thresholds in Table 2. The upper limit on the radius of the positive charge of gold is 33.5 fm (95% WALD 0.0 to 33.5 fm). These quantities and intervals appear in Figure 1, where the dashed rule marks the decision threshold.

UPPER LIMIT ON THE RADIUS OF THE POSITIVE CHARGE OF GOLD

33.5 fm

[0, 33] (95% WALD)

What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

At 150 degrees the counter recorded 94,117 particles where a diffuse atom predicts 119 — a Poisson excess of 306 standard errors. There is no version of this that is a fluctuation.

The hours went where the argument was weakest rather than where the information was densest, and that is what the bound is made of. Dropping the 150 degree witness and keeping every other station loses most of it.

Model checking

Model checking evaluates whether the observed counts and likelihood profiles are consistent with the model fit. A check of this kind can lower confidence in an estimate and can never establish one. These recorded diagnostics are reported in Table 3. Across the entire run, counts recorded across every station were 466849. At 150 degrees with the foil in, the count was 94117. In contrast, the counts a diffuse atom predicts at 150 degrees is 118.8. With the foil out, counts at 90 degrees were 11. In the profile likelihood analysis, the radius at which the profile likelihood bottoms is 32.3 fm. The radius the beam energy alone could never resolve past is 29.6 fm. These checks examine background levels, wide-angle counts, and profile minimums against resolution limits, leaving the remaining assumptions identified in earlier sections unexamined.

RADIUS AT WHICH THE PROFILE LIKELIHOOD BOTTOMS	RADIUS THE BEAM ENERGY ALONE COULD NEVER RESOLVE PAST	COUNTS AT 150 DEGREES, FOIL IN	COUNTS A DIFFUSE ATOM PREDICTS AT 150 DEGREES	COUNTS AT 90 DEGREES WITH THE FOIL OUT	COUNTS RECORDED ACROSS EVERY STATION
32.3 fm	29.6 fm	94,117	118.8	11	466,849

Discussion

The upper limit on the radius of the positive charge of gold settled below the threshold, on the favourable side. This indicates that the positive charge is concentrated in a centre far smaller than the atom itself. The estimate resolves the primary question in the direction favorable to a concentrated central charge. What these findings license, however, is bounded by how the effect is identified. The identification is downgraded via profile likelihood on the counts, through one forward step. The single-scattering assumption is asserted from the foil thickness rather than measured. In contrast, every other assumption the bound leans on was measured by a station the design bought for that purpose. Taking the bound as an empirical fact about the atom therefore depends on an assertion that was not directly verified in the run. The entire reading remains conditional on three unresolved assumptions, and what each one claims is stated in the limitations section. From this analysis, we learn that the count data place the charge radius below the threshold if the foil behaves as asserted. That result raises the question of whether single scattering can be verified by direct measurement rather than asserted from foil thickness.

Conclusions

The positive charge of a gold atom is concentrated in a centre far smaller than the atom itself. The upper limit on the radius of the positive charge of gold settles on the favourable side of the threshold at 33.5 fm (95% WALD 0.0 to 33.5 fm). Identification is downgraded. The result remains conditional on three unresolved assumptions detailed in the limitations section: single scattering, that the counter is linear below its rate cap, and that the two atoms differ only in the radius of the positive charge.

Limitations

Identification is downgraded, leaving three standing assumptions that do work the data does not do. The first assumption is single scattering, which asserts that a wide-angle count is one close encounter rather than several. This licenses treating each observed deflection as an isolated event. The claim would be challenged by a thicker foil, where two encounters would compound. The second assumption asserts that the counter is linear below its rate cap. Under this claim, every aperture is opened only as wide as keeps the rate under 1.5 counts per second, where the counter does not miss particles. This licenses the observed tallies across detector stations without adjustment for missed counts. The claim would be challenged by a station run wide enough to pile up, which would under-report the busiest angles. The third assumption asserts that the two atoms differ only in the radius of the positive charge. It claims that a hard centre and a charge spread through the atom are one model at two values of one number, the radius R of the positive charge. This licenses evaluating structural differences strictly through variations in spatial extent. The claim would be challenged by any difference between the two atoms that is not a matter of extent.

Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
Upper limit on the radius of the positive charge of gold	radius	33.5 fm (95% WALD 0.0 to 33.5 fm)	design.profile_likelihood on lam
Radius at which the profile likelihood bottoms	profile_min	32.3 fm	—
Radius the beam energy alone could never resolve past	floor	29.6 fm	—
Counts at 150 degrees, foil in	witness_counts	94117	—
Counts a diffuse atom predicts at 150 degrees	witness_if_diffuse	118.8	—
Counts at 90 degrees with the foil out	background_counts	11	—
Counts recorded across every station	total_counts	466849	—

Table 2. Quantities and their sources

Field	Value
estimator	profile likelihood on lam, one-sided
interval_mass	95%
observing_hours	200
seed	1911
world	nbs/case-studies/rutherford/scattering.py
evidence hash	e8f37b522ca34db4d0bfd8bebb94b492fca1ffcc356a5d ac330b4370d7de1efc

narration: abstract	gemini-3.5-flash-lite — verified
narration: title_page	gemini-3.7-flash — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 3. Run

Tables

Table 1. The design, as recorded parameters.

Step	Parameter	Value
Identification	route	profile likelihood on the counts, through one forward()
Identification	status	downgraded
The two atoms as one parameter	beam energy (MeV)	7.68
The two atoms as one parameter	closest approach D (fm)	59.25
The two atoms as one parameter	diffuse atom R (fm)	135000
The two atoms as one parameter	foil areal density (atoms/m ²)	2.36e+22
The two atoms as one parameter	hard centre R (fm)	7.3
Where to point the counter	anchor	1°, 2.5° (10 h)
Where to point the counter	background	90°, foil out (10 h)
Where to point the counter	bank	5°, 10°, 20°, 45° (45 h)
Where to point the counter	stations	9
Where to point the counter	witness	90°, 150° (135 h)
What shape to cut the aperture	beam rate (alphas/s)	2e+05
What shape to cut the aperture	rate cap (counts/s)	1.5
What shape to cut the aperture	widest aperture (sr)	0.2
How long at each	hours at 150°	110
How long at each	seed	1911
How long at each	total hours	200
Reading the bound off the counts	drop at 95% one-sided	1.921
Reading the bound off the counts	grid	41 points over lam in [-4, 4]
Reading the bound off the counts	nuisance parameters	log_a, log_c, log_w, log_b
Reading the bound off the counts	profiled parameter	lam

Table 2. Estimated quantities with their intervals and thresholds.

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
Upper limit on the radius of the positive charge of gold	33.5	fm	0.0 to 33.5 (95% WALD)	135000.0	below threshold

Table 3. Recorded diagnostics.

Check	Value	Unit	Interval
Radius at which the profile likelihood bottoms	32.3	fm	—
Radius the beam energy alone could never resolve past	29.6	fm	—
Counts at 150 degrees, foil in	94117	—	—
Counts a diffuse atom predicts at 150 degrees	118.8	—	—
Counts at 90 degrees with the foil out	11	—	—
Counts recorded across every station	466849	—	—

Figures

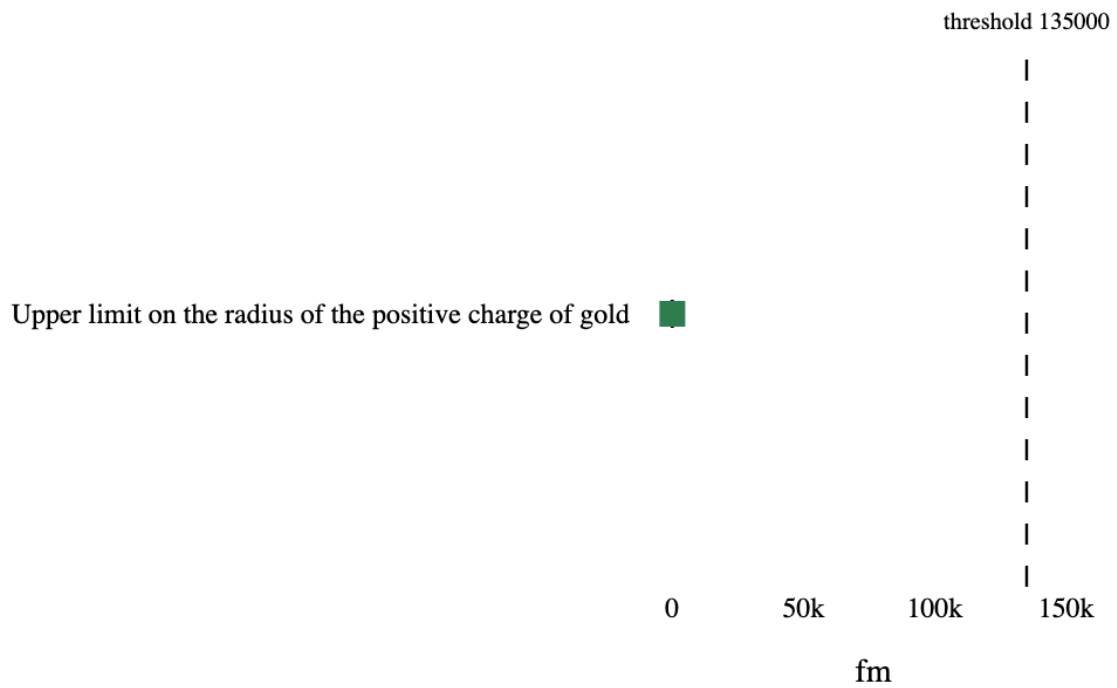


Figure 1. Estimated quantities with their intervals. The dashed rule marks the decision threshold; a row whose interval crosses it is unsettled rather than null.